Drip Guardia

Medi Sync

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM

About:

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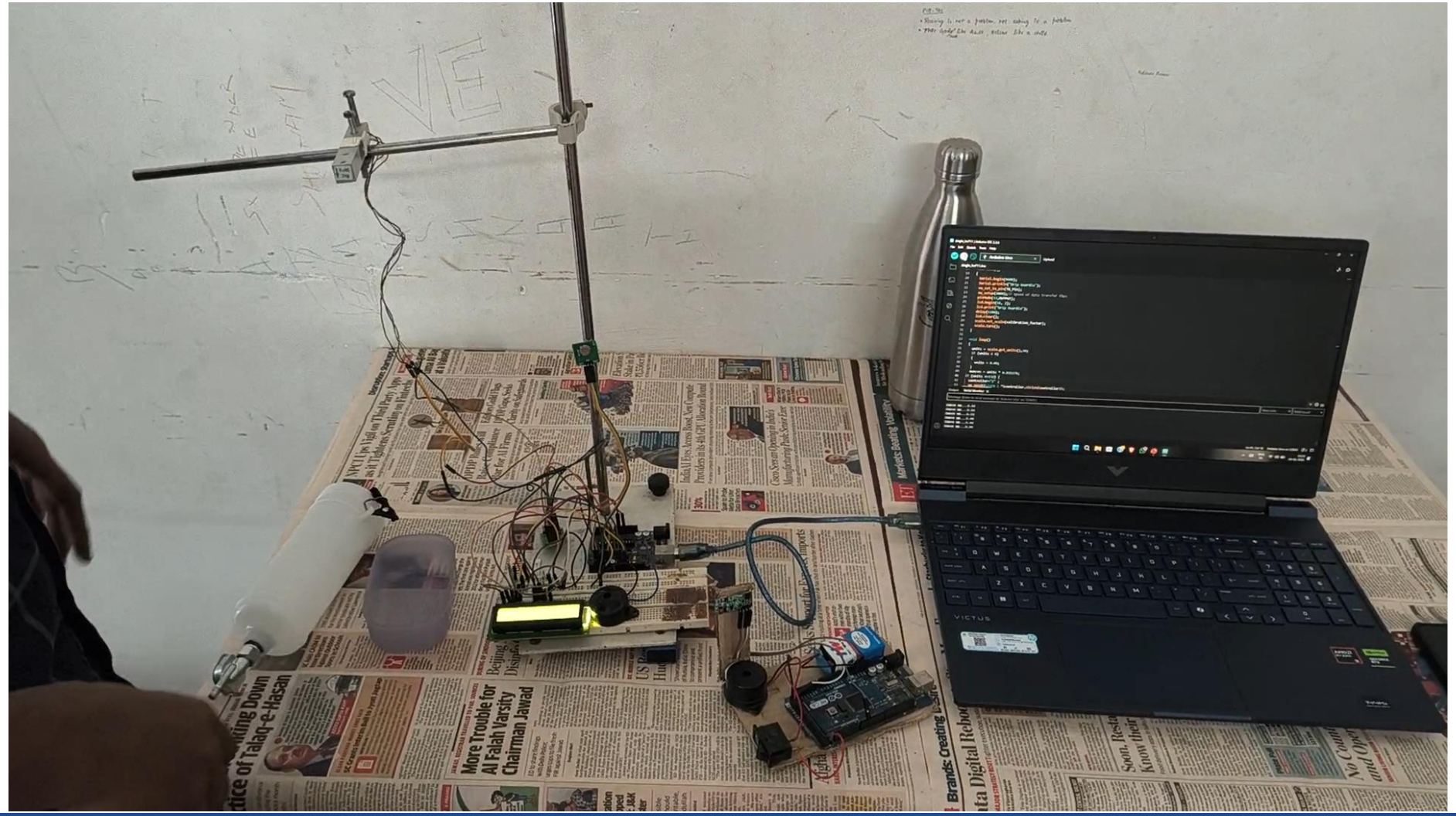
Problem Statement

- In Hospitals, nurses monitor the patients who are on drips to ensure the bottle (IV fluid bag) does not run empty.
- If it does, an empty IV fluid bag can force the air present in the bag into the IV tubing when the fluid runs out.
- This can potentially cause an air embolism.
- Finally, undetected dry IV bags can disrupt fluid flow into the patient, allowing blood to rise into the IV fluid bag.

Solution

- Our design is an Low-cost smart IV monitoring system to alert the nearby nurses and caretakers when the IV fluid is about to be completed in the IV bottle.
- It will measure the fluid weight to determine the amount of fluid in the IV bag.
- When the measured weight corresponds to 10 ml of fluid weight, the small alarm will ring until the IV bag is removed.
- There is a Liquid Crystal Display (LCD) which shows the live quantity of the IV fluid.
- There will also be an alarm in the nurse station to alert remote nurses, especially during the night.
- This design is highly portable and cost-efficient, as no changes are required to the existing drip stands and are installed directly to the drip stands.

Drip Guardia



Target Market

Market Need:

- Around 1.3 million hospital beds in India
- Around 24 million hospital beds worldwide
- Nearly 60% patients require IV fluid daily
- Hospitals need low-cost monitoring solution

Primary Customers:

- Hospitals (Government & Private)
- Multi-specialty Hospitals
- Small Clinics & Nursing Homes
- Intensive Care Units (ICU)
- Emergency Wards

Secondary Customers:

- Ambulance Services
- Home Healthcare Providers
- Old Age Homes & Care Centers
- Rural Health Centers
- Military Medical Units

Target Market

PRIMARY CUSTOMERS

- Hospitals (Government & Private) — 1 Million+
- Multi-specialty Hospitals — 500,000+
- Clinics & Nursing Homes — 400,000+
- ICU & Emergency Wards — 300,000+

SECONDARY CUSTOMERS

- Ambulance Services — 50,000+
- Home Healthcare Providers — 200,000+
- Old Age Homes — 20,000+
- Rural Health Centers — 20,000+

Market Need:

Around **1.3 Million** Hospital Beds in India

Around **24 Million** Hospital Beds Worldwide

Nearly **60%** Patients Require IV Fluid Daily

Around **1.3 Million** Hospital Beds in India

Business Model

TAM – Total Addressable Market:

- Estimated 20–24 million hospital beds globally
- Around 60% patients require IV fluids
- Potential users ≈ 12–14 million devices
- If price = ₹2,000 per device:
- TAM ≈ ₹24,000 – ₹28,000 crore

SAM – Serviceable Available Market:

- Initial focus on Indian hospital market
- India has about 1.18 million hospital beds in private hospitals alone
- Government hospitals add more than 8 lakh beds
- India needs millions of additional beds due to healthcare demand growth
- Total estimated beds ≈ 1.3 million
- 60% IV usage → 780k potential devices
- SAM ≈ ₹2400 crore

SOM – Serviceable Obtainable Market:

- Realistic market capture in first 3–5 years
- Assume capturing 1% of Indian market
- 1% of 1.2 million beds = 12,000 devices
- Revenue: 12,000 × ₹2,000 =
- SOM ≈ ₹2.4 crore

Competitive Analysis:

Existing Solutions:

- Infusion Pumps :
  - Accurate but very expensive
  - Needs trained staff
  - Not scalable for all beds
- One-Way (Anti-Reflux) Valves :
  - Prevents backflow
  - No monitoring or alerts
  - Passive solution

Drip Guardia Advantage:

- Real-time IV fluid monitoring
- Automatic alert system (local + remote)
- Low-cost & scalable
- Works with existing IV setups
- Reduces nurse workload

Feature	Infusion Pump	One-Way Valve	Drip Guardia
Cost	High	Low	Low
Monitoring	Yes	No	Yes
Alerts	Yes	No	Yes
Ease of Use	Moderate	Easy	Easy
Scalability	Low	High	High

Financial Projections:

Assumptions:

- Manufacturing cost per unit: ₹800 – ₹1,000
- Selling price: ₹1,800 – ₹2,500
- Profit per unit: ~₹1,000

Projected Scales:

Year 1 (Pilot Phase):

- Units sold: 500
- Revenue: ~₹10 Lakhs
- Profit: ~₹5 Lakhs

Year 2 (Expansion):

- Units sold: 2,000
- Revenue: ~₹40 Lakhs
- Profit: ~₹20 Lakhs

Year 3 (Scaling):

- Units sold: 5,000
- Revenue: ~₹1 Crore
- Profit: ~₹50 Lakhs

Future Roadmap — Drip Guardia

Phase 1:
Prototype Improvement
(0–3 months)

- Improve sensor accuracy
- Optimize circuit & battery
- Test alarm reliability
- Develop stable casing

Phase 2:
Testing & Validation
(3–6 months)

- Lab testing with IV fluids
- Accuracy for 10 ml alert
- Compare with manual checks
- Safety compliance

Phase 3:
Pilot Deployment
(6–9 months)

- Deploy 20–50 units
- Collect user feedback
- Usability improvements
- Optimize cost

Phase 4:
Product Optimization
(9–15 months)

- Add wireless alerts
- Enhance durability
- Simplify installation
- Manufacturing prep

Phase 5:
Scale & Market Entry
(15–24 months)

- Small-scale production
- Hospital partnerships
- Device certification
- Expand to new markets

Our Goal: ***Provide an affordable, reliable IV monitoring device that reduces nurse workload and improves patient safety in every hospital bed.***

Conclusion

- With this product, nurses or caretakers need not monitor the IV drips.
- Reduces nurse monitoring time by up to 30–40%.
- There will be no blood wastage, nor will the blood need to be sent again to the body (In case of a large amount of blood rising).
- It also allows the patients and the caretakers to take rest peacefully, especially at night.

Thank You